

[Internal] Network Security Audit Report

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Target: Local Area Network (Home Lab Environment)

Tools Used: Nmap, Wireshark

1. Executive Summary

The purpose of this audit was to identify active hosts, open ports, and potential vulnerabilities within a private network. This report outlines the discovery phase, packet analysis findings, and recommended security hardening steps.

2. Methodology

- Discovery:** Used **Nmap** for host discovery and service version detection.
- Packet Analysis:** Utilized **Wireshark** to monitor unencrypted traffic and evaluate protocol security.
- Risk Assessment:** Evaluated the findings against common cybersecurity threats (e.g., Man-in-the-Middle attacks, unauthorized access).

3. Key Findings

A. Network Mapping (Nmap)

Host IP	Status	Open Ports	Detected Service
192.168.1.1	Up	80, 443	Router Web Interface
192.168.1.15	Up	22, 445	SSH, SMB (File Sharing)

Observation: Port 445 (SMB) was found open on a workstation, which is a common vector for lateral movement in a network if not properly patched.

B. Traffic Analysis (Wireshark)

- Protocol Check:** Captured traffic showing active **HTTP** (unencrypted) requests to a local IoT device.
- Vulnerability:** Passwords or sensitive data sent via HTTP could be intercepted via a packet sniffer.

4. Proposed Mitigation & Hardening

- Close Unnecessary Ports:** Disable SMB (Port 445) on devices that do not require file sharing.
- Enforce Encryption:** Force the use of **HTTPS** for all local web-based management interfaces.

3. **SSH Hardening:** Update SSH to use Key-Based Authentication instead of passwords.

5. Conclusion

The network is currently functional but has minor security gaps that could be exploited by an internal threat. Implementing the above recommendations will significantly reduce the attack surface.